Economics of Education T4 2024/2025

Midterm – Grading Criteria (Points Total: 60)

Q1. 14 points

1.1. 4 points

Choose the correct statements about worldwide educational trends. Select all that apply.

Answer:

- [x] Education and basic skills have greatly improved in the past decades worldwide
- [x] On average, PISA test scores for Maths, Science, and Reading, are lower today than they were in the early 2000's for OECD countries
- [x] There is huge disparity of educational effectiveness across countries worldwide
- [ ] Girls are proportionally more enrolled in tertiary education in low-, middle-, and high-income countries

While girls are proportionally more represented in tertiary education in middle- and high-income countries, this trend is not observed in low-income countries. All the others are correct.

1.2. 10 points

While nearly every country has experienced gains in educational attainment over the last century, the timing and pace of this expansion have varied considerably.

a) Detail the historical patterns of basic education expansion in high-income and low-income countries. Provide at least two examples.

b) How have conflict and political regimes influenced the educational trajectories of different countries?

Answer:

Excellent (9–10 pts)

- Clearly explains that high-income countries expanded basic education first (early–mid 20th century) rather than middle- and low-income countries (late 20th century). Provides at least two well-chosen examples.
- Explicitly states that countries with conflicts or autocratic regimes suffer from reductions in basic education levels and data unavailability.
- Well-written, clearly organized.

Good (7–8 pts)

- Mostly accurate timeline and explanation. Provides two examples.
- Mentions that conflicts and lack of democracy negatively affect basic education levels.
- Mostly clear and organized.

Satisfactory (5–6 pts)

- General idea is present but lacks detail. Examples may be vague or underexplained
- The connection between education and conflicts/autocratic regimes is unclear or oversimplified.
- Disorganized or unclear writing

Fail (0–4 pts)

- Misunderstands or omits historical patterns. No examples or irrelevant ones
- Poorly written or hard to follow

Q2. 14 points

2.1. 4 points

Which of the following statements are true regarding studies that use monozygotic (MZ) twins to estimate the causal effect of education on earnings? Select all that apply.

Answer:

- [x] These studies can control for genetic endowments in the education-earnings association. ✓
- [x] These studies assume that differences in education between twins are exogenous. ✓
- [ ] Twin studies can eliminate all sources of bias in estimating the returns to education.
- [ ] Twin studies can provide an estimate of the causal impact of education on earnings.

While twin studies control for genetics and (at least some) shared family environment, they cannot account for all later-life factors or events that may influence both education and earnings. As such, it is not possible to fully claim causality based solely on this identification strategy.

2.2. 10 points

Describe the challenges faced in estimating the private benefits of education when trying to establish causality. In your response, identify and explain one strategy that you find convincing for addressing these challenges. Justify why this strategy is effective in overcoming the issues related to causality.

Answer:

Clarity and completeness of the description of challenges (4 points):
The response should accurately identify and describe the key challenges in establishing causality when estimating the private benefits of education. The student should mention issues such as selection, endogeneity, omitted variable bias, or reverse causality.

Explanation of a convincing strategy (3 points):
The student should explain at least one strategy used to address causality (e.g., minimum school leaving age, enrolment in school to avoid Vietnam lottery draft, minimum legal working age). The explanation should be clear and show a good understanding of how the strategy works.

Justification of why the strategy is convincing (3 points):
The student should provide a thoughtful and well-reasoned justification for why they believe the selected strategy is effective in addressing the causality challenges. This could include discussing how the strategy overcomes specific issues like confounding variables or reverse causality, and why it is appropriate for the context of estimating the private benefits of education.

9-10 points:

- **Challenges**: Clearly and accurately identifies and explains multiple challenges related to causality in estimating private benefits of education (e.g., endogeneity, omitted variables, reverse causality, measurement error). The description is detailed and insightful.
- **Strategy**: Effectively explains a relevant strategy to address these challenges (e.g., instrumental variables, randomized control trials, natural experiments). The explanation is thorough and demonstrates a deep understanding of the method.
- **Justification**: Provides a convincing, well-reasoned justification for why the strategy is effective. The justification includes strong reasoning based on theoretical or empirical support and directly addresses the specific causality challenges mentioned.

7-8 points:

- **Challenges**: Identifies and describes at least one key challenge but may miss or inadequately explain some nuances of the issue. The explanation is mostly correct but lacks depth.

- **Strategy**: Explains a relevant strategy, but the explanation is somewhat general or lacks clarity in certain areas. The student may demonstrate an understanding, but the application of the method could be more specific.
- **Justification**: Provides a reasonable justification but may be somewhat superficial or lack clear reasoning in some parts.

5-6 points:

- **Challenges**: Mentions a challenge, but the explanation is either incomplete or inaccurate. The student may describe the challenge too vaguely or with little connection to the question.
- **Strategy**: Explains a strategy but may misapply it, or the explanation may be too brief or unclear. The method described may not be the most convincing choice for addressing the causal issues mentioned.
- **Justification**: Provides a justification that is weak or unclear. The reasoning behind why the strategy works is either missing, underdeveloped, or not convincing in relation to the challenges described.

3-4 points:

- **Challenges**: Describes the challenge in a very basic or incorrect manner, lacking clarity or relevance to the concept of causality. There may be significant misunderstandings.
- **Strategy**: Mentions a strategy but fails to explain it clearly or accurately. The strategy might be misapplied or irrelevant.
- **Justification**: Provides little to no justification, or the justification is poorly reasoned, lacking connection to the challenges or strategy mentioned.

0-2 points:

- **Challenges**: No relevant description of challenges or a completely incorrect explanation.
- **Strategy**: No strategy provided or the strategy mentioned is irrelevant or wholly incorrect.
- **Justification**: No justification provided, or justification is incoherent or irrelevant to the question.

3. 6 points

3.1. 6 points

Answer:

There is no evidence that education has a positive causal impact on mental health.

Additionally, there is no evidence that education reduces fraud or cybercrime.

There is convincing causal evidence for all the other public benefits.

4. 14 points

4.1. 4 points

Which of the following statements are supported by empirical research on the relationship between education and civic engagement? Select all that apply.

Answer:

- [x] There is a positive association between education, civic engagement, and democratic outcomes. ✓
- [ ] While education and civic engagement are positively associated, there is no causal evidence linking them.
- [x] Countries with more equal educational outcomes tend to experience higher levels of democracy. ✓
- [x] Several studies document a causal link between education and civic engagement, including increased voting, political interest, tolerance, and participation. ✓

Many studies show a strong *positive correlation* between education levels and civic engagement. Research also does show that education equality (e.g., broader access) correlates with more robust democratic outcomes and with the strength of democratic institutions. Additionally, a number of causal studies find that more education increases civic behaviours like voting, political interest, tolerance, and participation.

4.2. 10 points

Consider slide 35 of class 4, that illustrates the political trade-off between education and civil engagement. Assume that you are in the presence of a predatory ruler.

Assume the following parameters:

- Wage: $w=3$
- Cost of education: $g=1$
- Probability that propaganda deceives the citizen about the ruler's type: $p=0.5$
- Cost of joining a political club: $c=0.5$
- Human capital productivity: $\alpha=2$
- Human capital: $h=2$
- Destruction of income due to protest: $\sigma=0.8$
- Probability that protest wins: $q=0.75$

Assume both the citizen and the ruler behave rationally. Answer the following:

1. What is the level of predatory taxation when the ruler provides education and the citizen remains politically inactive (PI)?

   Answer:
   A predatory ruler will tax all income since citizen can't protest.

   Total wage = $w + \alpha*h = 3 + 2*2 = 7$

   2.5 points if answer is 7; 0 any other number

2. What is the level of predatory taxation when the ruler provides education and the citizen becomes politically active (PA)? (assume $\sigma < 1$)

   Answer:
   A predatory ruler will tax as much as possible without making citizen protest. Citizen is indifferent between protest or not if:

$q*\sigma(w + \alpha h - \alpha c) = w + \alpha h - \alpha c - t_{c=1}$

Solving for $t_{c=1}$ yields:

$t_{c=1} = (1 - q*\sigma)(w + \alpha h - \alpha c)$

One can show that this is always larger than risking protest:

$(1-q\sigma)(w + \alpha h - \alpha c) > (1-q)\sigma(w + \alpha h - \alpha c)$

$(1-q\sigma) > (1-q)\sigma$

$1 > \sigma - q\sigma + q\sigma$

$1 > \sigma$

For any positive destruction, this is, $\sigma < 1$

So, tax = (1- 0.75*0.8)(3+2*2 -2*0.5) = 0.4*6=2.4

2.5 points if answer is 2.4, 0 if any other number

3. What is the level of predatory taxation when the ruler does not provide education (NE)?

   Answer:
   $t_{e=0} = w = 3$

   2 points if answer is 3, 0 if any other number

4. Based on the three possible outcomes above, will the ruler choose to provide education? What are the expected rents in each case (provides education vs does not provide education)?

   Answer:
   Ruler wants to maximize rent extraction. In the first case, the maximum rent it can extract is wage minus cost of education: 7 – 1 = 6, in the second case, the maximum rent is 2.4 -1 = 1.4. In the third case, the maximum rent it can extract is 3 (no cost of education)

   If it provides education and it implements propaganda, then there is probability p=0.5 that the citizen believes ruler is benevolent in that case it will choose to not join a club. Otherwise, it learns about its true type, and joins a club. This means that the maximum rent in expectation if it provides education is 0.5*6 + 0.5*1.4 = 3+0.7 = 3.7

   If it doesn't provide education then it can have rents = 3, which is lower.

   Yes, the ruler chooses to provide education, if it provides education, expected rent is 3.7, if it doesn't, the rent is 3.

- **3 points:**
  The answer correctly describes the ruler's revenue (rent) in each scenario — with and without education — including the cost of education. The student correctly concludes that the ruler will provide education because the expected revenue with education is higher than without.
- **1.5 points:**
  The student correctly calculates the tax rates but forgets to subtract the cost of education to the rents, incorrectly concluding that the ruler earns 4.7 from education and thus provides it.
- **0 points:**
  The student calculates any other rent values.

Q5. 12 points

5.1. 6 points

Consider a labour market where there are two types of workers: high productivity and low productivity.

The minimum wage set by the government is 30.000eur.

The employer offers a basic earnings premium of 20.000eur if the employee has a degree. Additionally, there are several types of nationalities, and an employer discriminates against some, offering an additional education-wage premium of 10.000 to European individuals.

The cost of education is 10.000 for a high productivity individual, and 15.000 for a low productivity individual.

Assume 3 time periods, with education lasting 1 period, and working lasting 2. Assume an interest rate of 2%.

How much will a non-European individual earn in present value, if it chooses to pursue education? And if it doesn't? Will the individual pursue education? Assume payments are made in the start of the period.

Answer:

The individual will not pursue education since the NPV without education is 88,247 and with education is 79,725.

The individual will get a minimum wage of 30.000 if it doesn't choose to pursue education. This means it will get 30k + 30k/1.02 + 30k/(1.02*1.02) = 88,247

If it chooses to pursue education if will get -10000 + 50000/1.02 + 50000/(1.02*1.02) = 87,078

The individual will not pursue education as the PV of no education is higher

6 points – correctly calculates PVs and correctly predicts that the individual will not receive education.

3 points – correctly calculates one PV.

OR

correctly calculates both PV assuming that payments are made at the end of the period:

30000/1.02 + 30000/(1.02*1.02) + 30000/(1.02*1.02*1.02) = 86,516

-10000/1.02 + 50000/(1.02*1.02) + 50000/(1.02*1.02*1.02) = 85,371

1.5 points - correctly calculates one PV, assuming that payments are made at the end of the period.

0 points – any other.

5.2. 4 points

Assume that education increases skills and productivity. A policymaker is attempting to maximize productivity, and as such, is interested in making sure that all high productivity individuals obtain a degree. Which policy could achieve that? Select all that apply.

Answer:

- [ ] Increasing the minimum wage
- [x] Decreasing the minimum wage ✓
- [ ] Making sure that there can't be wage differences between ethnicities
- [x] Establishing a minimum wage for degree holders ✓

If the minimum wage goes up, then the baseline (no education) option becomes more attractive. That means education becomes relatively less attractive, because the alternative is better paid. So fewer high productivity individuals will choose to educate themselves.

Making sure that there can't be wage differences between countries of origin is an important equity policy, but its effect on education incentives for high productivity individuals is unclear. It is possible that the employer lower the earnings premium for European individuals, resulting in a situation where no one gets a degree.

Decreasing the minimum wage, would make the no-education route less attractive, such that high productivity individuals are more likely to find education worthwhile. This policy could help achieve the goal.

Establishing a minimum wage for degree holders would increase the return on education, making it more attractive. This could also helps achieve the goal.

5.3. 2 points

In this setting, if your goal is to both maximize productivity and promote social justice, which policy would you choose? Briefly explain your reasoning.

Answer:

- **2 points** – Student correctly identifies Option D ("Establishing a minimum wage for degree holders") *and* explains that this is the only option that can both guarantee that high productivity individuals pursue a degree and that guarantees a certain level of return on investment for all individuals, regardless of their ethnicity.

- **0 points** – Student picks an incorrect option (e.g., A or B) or picks correct option but fails to provide a valid explanation *or* gives no explanation at all.